```c
#include <kipr/wombat.h>

//Sensorfunctions
void findLine();
void forward();
void backwards();
void findWall();

//Pomfunctions
void firstpoms2();
void firstpoms2return();

//Armfunctions
void ArmsDown();
void ArmsUp();
void rightArmReturn();
void rightArmD();
void leftArmD();
void leftArmReturn();

int rightSensFor = 0;
int backLine = 3;
int lmotor = 0;
int rmotor = 3;
int bottomLine = 2000;
int bottomLine2WHAT = 2000;
int speed = 90;
int posl = 0;
int speedservo = 20;

//Servo ports
int pomservo = 1;
int servoright = 0;
int servoleft = 3;

//LiftArmServoVars
int liftUpLeft = 100;
int liftUpRight = 1900;
int midupLeft = 738;
int midupRight = 1257;
int midLeft = 710;
int midRight = 1278;
int liftDownRight = 100;
int liftDownLeft = 1900;
```

```
//Pomservo vars
int pomservoout = 250;
int pomservoin = 1458;

int main()
{
    enable_servos();
    findLine();
    firstpoms2();
    firstpoms2return();
    forward();
    msleep(1000);
    findLine();
    firstpoms2();
    firstpoms2return();
    findWall();
    printf("function fired parent\n");
    ArmsDown();
        //msleep(10000);
    msleep(100);
    backwards();
    msleep(1000);
    ArmsUp();

    disable_servos();
    return 0;
}
//Pom Servo
void firstpoms2return(){


    int pos = pomservoout;
    while(pos < pomservoin){
        set_servo_position(pomservo, pos);
        msleep(20);
        pos +=20;
    }


    disable_servos();

}
```

```
void firstpoms2(){
   enable_servos();
   int pos = pomservoin;
   while(pos > pomservoout){
       set_servo_position(pomservo, pos);
      msleep(20);
      pos -=20;
   }

   disable_servos();

}

//ArmServos

void ArmsDown(){
         enable_servos();
   set_servo_position(servoright, midupRight);
   set_servo_position(servoleft, midupLeft);
   msleep(1000);
   set_servo_position(servoright, midRight);
   set_servo_position(servoleft, midLeft);
   msleep(1000);
   set_servo_position(servoright, liftDownRight);
   set_servo_position(servoleft, liftDownLeft);
   msleep(1000);
}
void ArmsUp(){
         enable_servos();

   set_servo_position(servoright, midRight);
   set_servo_position(servoleft, midLeft);
   msleep(1000);
   set_servo_position(servoright, midupRight);
   set_servo_position(servoleft, midupLeft);
   msleep(1000);
   set_servo_position(servoright, liftUpRight);
   set_servo_position(servoleft, liftUpLeft);
   msleep(1000);
}


//Sensor Movement Functions
void findLine()
```

```
{
    // At start, look for line then stop for a second
    while(analog(rightSensFor) <= bottomLine){
        forward();

    }
    ao();
}

void findWall()
{
    // At Line, look for line then stop for a second
    while(analog(backLine) <= bottomLine){
        forward();

    }
    ao();
}
//Movement
void forward()
{
    printf("I'm going forward\n");

    motor(lmotor,speed);
    motor(rmotor,speed);
}

void backwards()
{
    printf("I'm going backwards\n");

    motor(lmotor,-speed);
    motor(rmotor,-speed);
}
```